

EXAM 7 – SPRING 2012

7. (3 points)

For a set of retrospectively rated workers compensation policies, the following factors apply:

- Basic premium factor: 0.20
- Loss conversion factor: 1.25
- Tax multiplier: 1.04

There will be three retrospective adjustments, with the following estimated factors:

Retro Adjustment	% Loss	Loss
<u>Period</u>	<u>Emerged</u>	<u>Capping Ratio</u>
First	75.1%	0.93
Second	19.5%	0.72
Third	5.4%	0.55

Also assume:

- Total expected unlimited loss for these policies is \$205,000,000.
- The expected loss ratio is 75%.
- Total current booked premium is \$220,000,000.
- None of the policies has yet had its first retro adjustment.

Estimate the retrospective premium asset for this set of policies.

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Question 7 Sample Answer

Solution 1

Since no adjustments have been made, find the cumulative capping ratio.

$$\frac{CL_{UH}}{L_{UH}} = .751(.93) + .195(.72) + .054(0.55) = .869$$

$$L_{UH} = 205M \text{ (given)} \Rightarrow CL_{UH} = 205M(.869) = 178.05M$$

$$\text{Standard premium} = 205M / .75 = 273.33M$$

$$\begin{aligned} \text{Premium Asset} &= [0.2(273.33M) + (178.05M)(1.25)](1.04) - 220M \\ &= 68,316,578 \end{aligned}$$

Solution 2

<u>Retro Period</u>	<u>%Emerg'd × 205M</u> <u>Total loss in period</u>	<u>(Total loss in period)×(LCF)</u> <u>Capped loss</u>	
1	153,955,000	143,178,150	←153,955,000×0.93
2	39,975,000	28,782,000	
3+	11,070,000	<u>6,088,500</u>	
		178,048,650	

$$\begin{aligned} \text{Initial} \quad \text{Prem} &= 205,000,000 / 0.75 = \\ \text{BP} &= \text{basic prem } 205M/0.75 = 54,666,667 \\ \text{Prem} &= (\text{BP} + \text{Capped Loss} \times \text{LCF}) \times \text{TM} \\ \text{Premium} &= (54,666,667 + 178,048,650 \times 1.25) \times 1.04 \\ &\text{including adjustments} \\ &288,316,579 \end{aligned}$$

$$\begin{aligned} \text{Premium asset} &= (\text{estimated premium including adjustments}) - \text{booked premium} \\ &68,316,579 \end{aligned}$$

Solution 3

$$\begin{aligned} PDLD_1 &= \left[\frac{BP \times TM}{SP \times \%Loss \times ELR} \right] + \left[\frac{CL_1}{L_1} \times LCF \times TM \right] \\ &= \left[\frac{(0.2)(1.04)}{(.751)(.75)} \right] + [(93) \times 1.25 \times 1.04] = 1.578 \end{aligned}$$

$$PDLD_2 = \frac{CL_1 - CL_2}{L_2 - L_1} \times LCF \times TM = 0.72 \times 1.25 \times 1.04 = 0.936$$

$$PDLD_3 = 0.55 \times 1.25 \times 1.04 = 0.715$$

$$CPDLD = \frac{(1.578)(.751) + (.936)(.195) + (.715)(.054)}{(.751) + (.195) + (.054)} = 1.406$$

$$\text{Asset} = (205M \times 1.406) - 220M = \$68M$$

Examiner Comment

Many candidates interpreted the loss capping ratios to be cumulative rather than incremental, producing nonsensical results. Candidates should have recognized that additional loss dollars will not reduce the premium as implied in this approach.

A number of candidates calculated estimated premium incorrectly by applying the CPDLD to loss dollars at each period and summing.

Many candidates incorrectly calculated the basic premium.